

M12 Circular Sensor/Actuator Connector

SKU / Model Number: M12-8P-XCODE-F
Manufacturer: SensorConnect GmbH
Category: Connectors | ETIM Code: EC001030

PRODUCT OVERVIEW

The M12 X-coded 8-pin connector is designed for high-speed Industrial Ethernet (10 Gbit/s Cat6A) in factory automation. Supports PROFINET, EtherCAT, and EtherNet/IP. Shielded 360-degree for EMC compliance.

TECHNICAL SPECIFICATIONS

Coding	X-coded (Cat 6A)
Pin Count	8 (4 pairs)
Data Rate	Up to 10 Gbit/s
Rated Current	2 A per contact
Rated Voltage	60 V
Mating Cycles	> 100
Cable Category	Cat 6A (IEC 61156-6)

ELECTRICAL RATINGS

Characteristic Impedance	100 ohm +/- 10%
Insertion Loss (500 MHz)	< 0.2 dB
Return Loss (500 MHz)	> 20 dB
Crosstalk (NEXT, 500 MHz)	> 39 dB
Shielding	360 deg, > 60 dB (up to 3 GHz)

DIMENSIONS & MECHANICAL

Thread	M12 x 1
Connector Diameter	15.8 mm
Length (cable version)	48 mm
Cable OD	5.5 to 9.0 mm
Weight	25 g

OPERATING CONDITIONS

Temperature Range	-25 to 85 deg C
IP Rating	IP65 / IP67 (mated)
Vibration	IEC 60068-2-6 (5g)
Shock	IEC 60068-2-27 (30g)
Salt Spray	96 hours

MATERIALS OF CONSTRUCTION

Shell	Zinc alloy, nickel plated
Insulator	PBT, UL94 V-0
Contacts	CuSn, gold plated (0.75 um)
Cable	PUR jacket, halogen-free
Strain Relief	PBT with cable clamp

PERFORMANCE DATA / SELECTION TABLE

Coding	Pins	Data Rate	Application
A-coded	4/5	100 Mbit/s	Sensor/Actuator
B-coded	5	100 Mbit/s	PROFIBUS
D-coded	4	100 Mbit/s	Ethernet 100BASE-TX
X-coded	8	10 Gbit/s	Ethernet Cat 6A

CERTIFICATIONS & APPROVALS

IEC 61076-2-109 | CE | UL | PROFINET V2.4 | EtherCAT conformance | RoHS

Note: Specific certification documents and test reports are available upon request. All products are manufactured under ISO 9001:2015 quality management system.

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